

FDP LAB : -01

ROS 2 Installation, Workspace & Package Creation

and Useful Commands

ROS2 Humble Install link -

<https://docs.ros.org/en/humble/Installation/Ubuntu-Install-Debs.html>

Step for Creating Workspace -

```
mkdir ros2_ws(ros2_ws is workspace name)
cd ros2_ws
mkdir src
colcon build
```

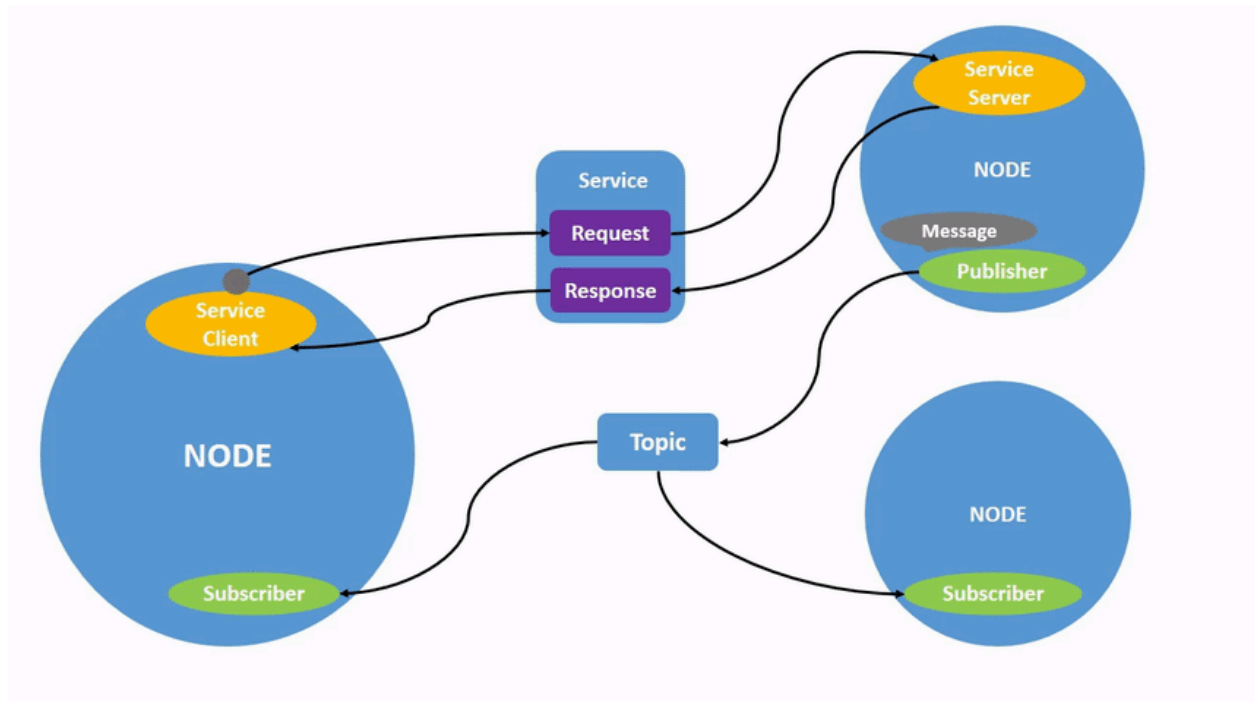
Steps for creating Python package

```
cd ~/ros2_ws/src
ros2 pkg create --build-type ament_python my_python_pkg
```

Steps for creating a C++ package

```
cd ~/ros2_ws/src
ros2 pkg create --build-type ament_cmake my_cpp_pkg
```

Node and Topic



Important Command for ros2 node

ros2 node list: This command lists all currently running nodes.

```
ubuntu@ubuntu-VirtualBox:~$ ros2 node list
/teleop_turtle
/turtlesim
```

ros2 node info <node_name>: This command provides details about a specific running node.

```

ubuntu@ubuntu-VirtualBox:~$ ros2 node info /turtlesim
/turtlesim
Subscribers:
  /parameter_events: rcl_interfaces/msg/ParameterEvent
  /turtle1/cmd_vel: geometry_msgs/msg/Twist
Publishers:
  /parameter_events: rcl_interfaces/msg/ParameterEvent
  /rosout: rcl_interfaces/msg/Log
  /turtle1/color_sensor: turtlesim/msg/Color
  /turtle1/pose: turtlesim/msg/Pose
Service Servers:
  /clear: std_srvs/srv/Empty
  /kill: turtlesim/srv/Kill
  /reset: std_srvs/srv/Empty
  /spawn: turtlesim/srv/Spawn
  /turtle1/set_pen: turtlesim/srv/SetPen
  /turtle1/teleport_absolute: turtlesim/srv/TeleportAbsolute
  /turtle1/teleport_relative: turtlesim/srv/TeleportRelative
  /turtlesim/describe_parameters: rcl_interfaces/srv/DescribeParameters
  /turtlesim/get_parameter_types: rcl_interfaces/srv/GetParameterTypes
  /turtlesim/get_parameters: rcl_interfaces/srv/GetParameters
  /turtlesim/list_parameters: rcl_interfaces/srv/ListParameters
  /turtlesim/set_parameters: rcl_interfaces/srv/SetParameters
  /turtlesim/set_parameters_atomically: rcl_interfaces/srv/SetParametersAtomically
Service Clients:
Action Servers:
  /turtle1/rotate_absolute: turtlesim/action/RotateAbsolute
Action Clients:

```

Important Command for ros2 topic

ros2 topic list: Lists all available topics.

```

ubuntu@ubuntu-VirtualBox:~$ ros2 topic list
/parameter_events
/rosout
/turtle1/cmd_vel
/turtle1/color_sensor
/turtle1/pose

```

ros2 topic info <topic_name>: Shows details about a specific topic.

```

ubuntu@ubuntu-VirtualBox:~$ ros2 topic info /turtle1/cmd_vel
Type: geometry_msgs/msg/Twist
Publisher count: 0
Subscription count: 1

```

ros2 topic echo <topic_name>: Prints messages published to a topic.

```

ubuntu@ubuntu-VirtualBox:~$ ros2 topic echo /turtle1/cmd_vel
linear:
  x: 2.0
  y: 0.0
  z: 0.0
angular:
  x: 0.0
  y: 0.0
  z: 0.0
---
linear:
  x: 2.0
  y: 0.0
  z: 0.0
angular:
  x: 0.0
  y: 0.0
  z: 0.0
---

```

ros2 topic pub <topic_name> <message_type> (with additional arguments):
Publishes a message to a topic.

```
ubuntu@ubuntu-VirtualBox:~$ ros2 topic pub --once /turtle1/cmd_vel geometry_msgs/msg/Twist "{linear: {x: 2.0, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 1.8}}"
```

publisher: beginning loop
publishing #1: geometry_msgs.msg.Twist(linear=geometry_msgs.msg.Vector3(x=2.0, y=0.0, z=0.0), angular=geometry_msgs.msg.Vector3(x=0.0, y=0.0, z=1.8))

ROS 2 Build and Execution Commands

```
ros2 run <package> <executable>
```

Run a node

```
ros2 launch <package> <file>
```

Launch a launch file

```
colcon build
```

Build all packages in a workspace

```
source install/setup.bash
```

Source the built workspace
